28/11/2025 – Matematicas Discretas 1 (UdeA)

→ Apuntes antiguos:

1. Introducción:

https://www.ted.com/talks/jose_americano_n_l_f_de_freitas_how_exactly_does_binary_code_work

2. Informacion

Claude Shannon

world — señales →

contexto

Representacion
(Datos)

Datos

Numericos

23.8

76

Alfanumericos
(letras, caracteres especiales, caracteres numericos) ...

c a t

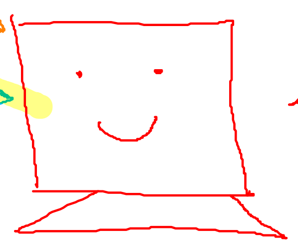
Sistemas numericos

Codigo Ascii

Representacion

(Mapeo = Función)

0 1 0 1 0 1 0



3. Sistemas numericos.

Sistemas numericos posicionales

2

D U
23

8.35

1'000000

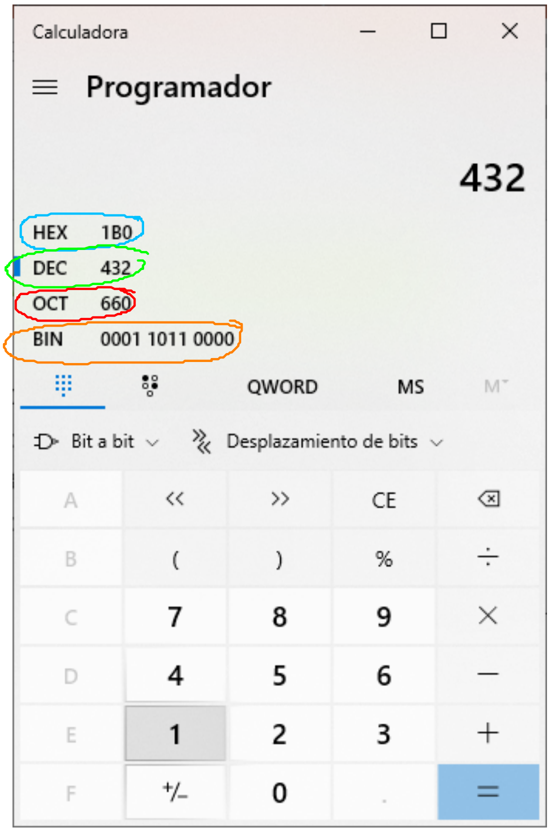
$N = d_n \ldots d_1 d_0 . d_{-1} d_{-2} \ldots d_{-m}$

Numero : N

$$\overset{D}{2}\overset{U}{3} = 2 \times \underset{10^1}{10} + 3 \times \underset{10^0}{1}$$

$$d_1 \, d_0 = d_1 \times 10^1 + d_0 \times 10^0$$

$$N_b = \sum_{i=-m}^{n} d_i \cdot b^i$$



$b = 10$

$N = 432$

432_{10}

$b = 2$

$0001\,10110000_2$

$b = 8$

660_8

$b = 16$

$1B0_{16}$

$$\overset{1\;0\;-1}{25.4_6} = 2(6^1) + 5(6^0) + 4(6^{-1})$$

$b = 6$

$$6^{-1} = \frac{1}{6} =$$

$$= 2(6) + 5(1) + 4(0.17)$$

$$= 12 + 5 + 0.68$$

$$= 17.68_{10}$$

Base m $\xrightarrow{\text{Expanción}}$ Base 10

3. Sistemas numericos $\underbrace{\text{Decimal}}_{b=10}$, $\underbrace{\text{binario}}_{b=2}$, $\underbrace{\text{octal}}_{b=8}$ y $\underbrace{\text{hexadecimal}}_{b=16}$

Decimal:

- base: $b = 10$
- Digitos: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- Ejemplos: 23.6_{10}

  123_{10}

  666

  -7

Binario:

- base: $b=2$
- Digitos: 0, 1

X $0 1 \cancel{2} \cancel{3}_2$ $0110 0001_2$

11.01_2

Octal:

- base: $b=8$
- Digitos: 0, 1, 2, 3, 4, 5, 6, 7

1203_8

X $1\cancel{8}_8$

Hexadecimal:

- base: $b = 16$
- Digitos: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, $\underset{(10)}{A}$, $\underset{(11)}{B}$, $\underset{(12)}{C}$, $\underset{(13)}{D}$, $\underset{(14)}{E}$, $\underset{(15)}{F}$

$12E_{16}$ X $\cancel{Z}A3_{16}$

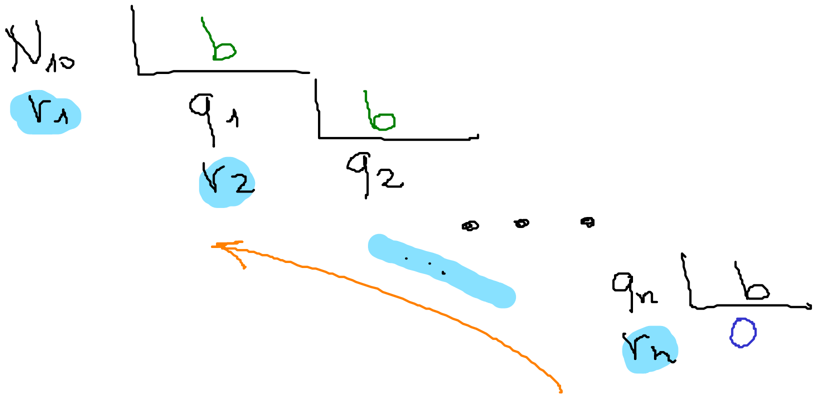
$12FF_{16}$

$$1AD_{16} = 1(16^2) + A(16^1) + D(16^0)$$
$$= 1(16^2) + 10(16^1) + 13(16^0)$$
$$= 1(256) + 10(16) + 13(1)$$
$$= 429_{10}$$

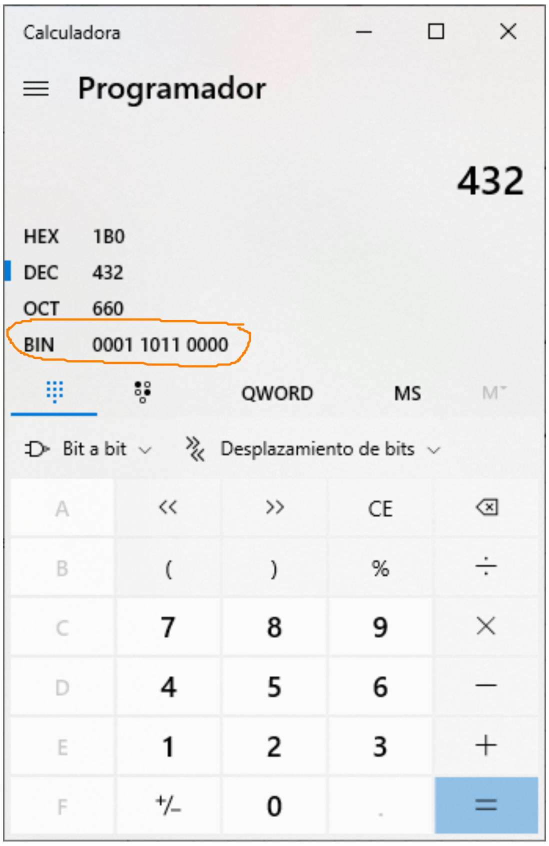
$b = 16$

Como hago

4. Divisiones sucesivas



$$N_b = r_n r_{n-1} \ldots r_1$$



$N = 432_{10}$

1. $N_2 = ?$

✓ $N_2 = 110110000$

2. $N_8 = ?$

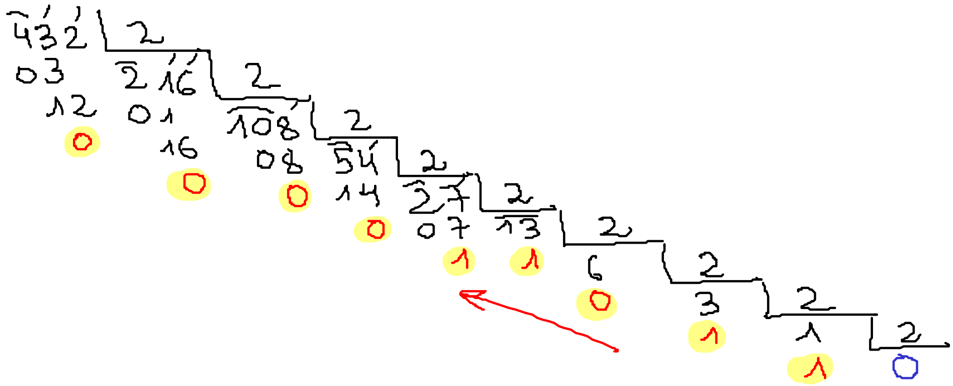
$N_8 = 660_8$

2. $N_{16} = ?$

$N_{16} = 1B0_{16}$

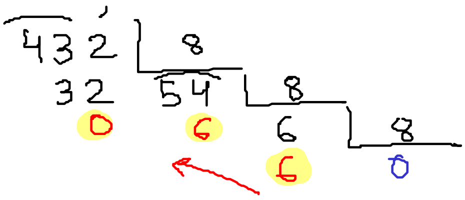
$N_{10} = 432 = 432_{10} \longrightarrow N_2 = ?$

$b = 2$



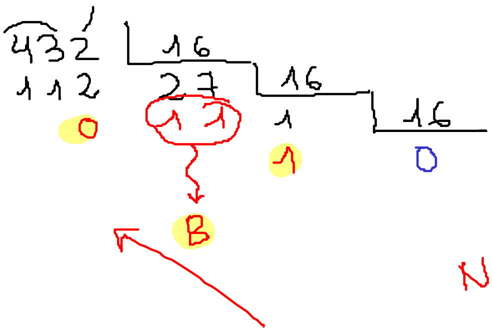
$N_2 = 110110000_2$

$N_{10} = 432 = 432_{10} \longrightarrow N_8 = ?$



$N_8 = 660_8$

$N_{10} = 432 = 432_{10} \longrightarrow N_{16} = ?$

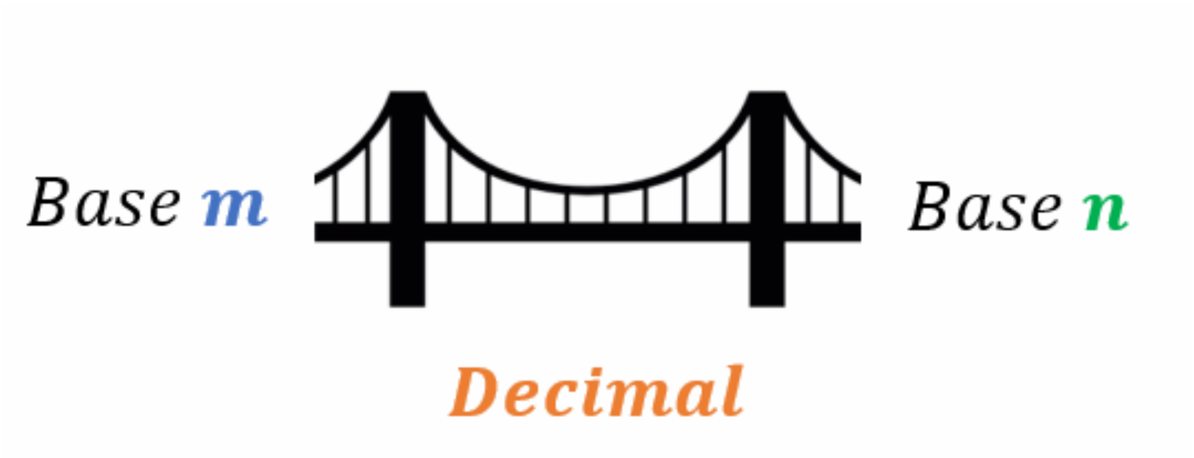


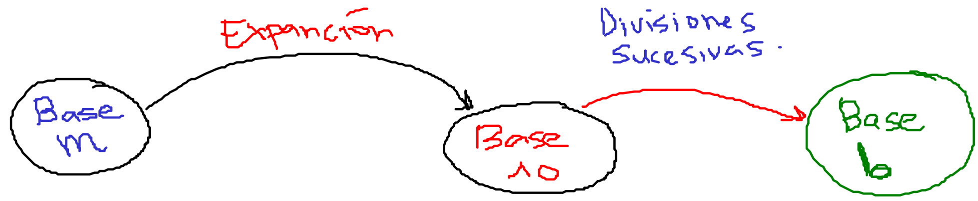
$N_{16} = 1B0_{16}$

$m \neq 10$

$n \neq 10$

5. Método del Puente





Ejemplo: $42_5 \longrightarrow N_9 = ?$

P1. $42_5 \longrightarrow N_{10}$:

$$42_5 = 4(5^1) + 2(5^0)$$
$$= 4(5) + 2(1)$$
$$= 20 + 2$$
$$= 22$$

$$42_5 = 22_{10}$$

P2. $22_{10} \longrightarrow N_9 = ?$

$$\begin{array}{r|l} 22 & 9 \\ \hline 4 & 2 \;|\; 9 \\ & 2 \quad 0 \end{array}$$

$$22_{10} = 24_9$$

$$42_5 = 24_9$$

6. Metodo Directo (Binaria, octal, Hexadecimal)

$b = 2$ $\quad b = 8 = 2^3$ $\quad b = 16 = 2^4$

$$\begin{array}{l} 7 \,|\, 2 \\ 1 \quad 3 \,|\, 2 \\ \quad\;\; 1 \quad 1 \,|\, 2 \\ \qquad\quad\;\; 1 \quad 0 \end{array}$$

$$7 = 111_2$$

$$\begin{array}{l} 12 \,|\, 2 \\ 0 \quad 6 \,|\, 2 \\ \quad\;\; 0 \quad 3 \,|\, 2 \\ \qquad\quad\;\; 1 \quad 1 \,|\, 2 \\ \qquad\qquad\quad 1 \quad 0 \end{array}$$

$$12 = 1100_2$$

Octal	Binario
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Hexadecimal	Binario
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A(10)	1010
B(11)	1011
C(12)	1100
D(13)	1101
E(14)	1110
F(15)	1111

Ejemplo:

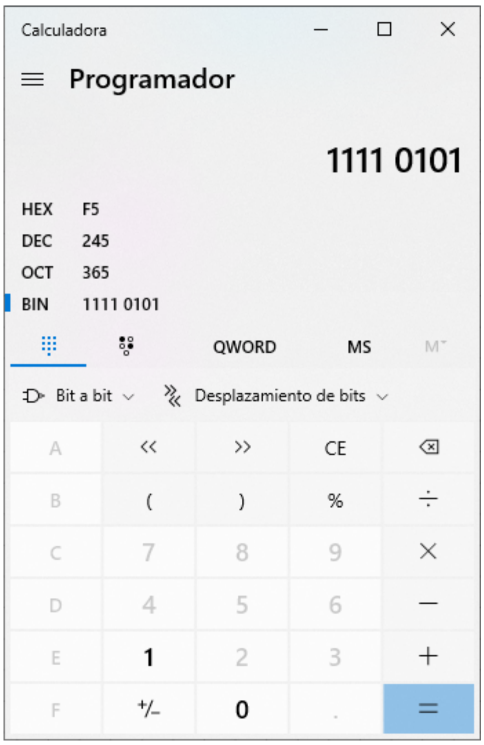
$$175_8 \longrightarrow N_2 = ?$$

$$175_8 = \underset{001}{1} \quad \underset{111}{7} \quad \underset{101}{5} = \underbrace{001}_{1}\,\underbrace{111}_{7}\,\underbrace{101}_{5}{}_2$$

$$00011110101_2 = N_{16} = ?$$

$$b=8 \quad \underbrace{000}_{0}\,\underbrace{011}_{3}\,\underbrace{110}_{6}\,\underbrace{101}_{5} = 365_8$$

$$b=16 \quad \underbrace{0000}_{0}\,\underbrace{1111}_{F}\,\underbrace{0101}_{5} = F5_{16}$$



$$1A_{16} \longrightarrow N_8 = ?$$

P1. $1A_{16} \longrightarrow N_2$

$$1A_{16} = \underset{0001}{1} \quad \underset{1010}{A} = \cancel{000}11010 = 11010_2$$

P2. $N_2 \longrightarrow N_8$

$b=8$ $\quad \underbrace{011}_{3}\,\underbrace{010}_{2} = 32_8$

$$\boxed{1A_{16} = 32_8}$$

Apuntes nuevos (agregados el 28/11/2025)